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An association rule-based approach for frequent item mining of multi-stage access data

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Abstract
The processing of large-scale datasets is complex and requires high efficiency. The database needs to be scanned multiple times by traditional Apriori algorithms to generate candidate itemsets, resulting in significantly reduced efficiency, but also have too simple a support threshold setting, which fails to fully consider the importance and differences of different itemsets in practical business, thereby affecting the accuracy of frequent item mining. Therefore, a multi-stage frequent item mining method for accessing data based on association rules is proposed. By applying the concepts of clustering analysis and Boolean vector operation, the Apriori algorithm for association rules is improved. The matrix is compressed through clustering and counting of the same transactions to reduce the size of multi-stage database access. Only one database scan is required, and frequent itemsets are directly generated without generating candidate itemsets, greatly improving the efficiency of the algorithm; Introducing weights into the improved Apriori algorithm and designing a weighted support function, by assigning different weights to different itemsets, more accurately reflects the importance and differences of itemsets in practical applications, thereby improving the accuracy of frequent item mining. Experiments have shown that this method can effectively mine frequent items in multi-stage access data, and the Pearson coefficient of frequent items mined by this method is close to 1, that is, the mining accuracy of frequent items is high; at different minimum weighted supports, this method has multi-stage comprehensive evaluation function value of mining frequent items in access data is higher, that is, the mining effect of frequent items is better.

Keywords Association rules, Multi-stage, Access data, Frequent item mining, Apriori algorithm, Boolean vector

1 Introduction

With the explosive growth of data scale, data mining technology has become a necessary tool for processing large-scale data [1]. Among many data mining techniques, frequent item mining is an important method to find frequently occurring item sets or patterns in data sets [2]. Multi-stage access data frequent item mining is a data mining technology [3] that finds frequently appearing item sets or patterns in the multi-stage data processing process. As the complexity of the data processing process and the size of the data increase, multi-stage access data frequent item mining is becoming more and more widely used in various fields [4]. The multi-stage access data frequent item mining method can be applied to many scenarios, such as e-commerce, social networks, finance, etc. In these scenarios, data is usually divided into multiple stages for processing [5], such as users browsing products, placing orders, paying, etc. By looking for frequently occurring item sets or patterns in each stage, it can help companies better understand user behavior and market trends, thereby formulating more effective business strategies. The multi-stage access data frequent item mining method has many advantages: First, it can discover interesting relationships and patterns between different stages, thereby better understanding the distribution and trends of the data [6]. Secondly, it can handle large-scale data sets with high efficiency and performance. Finally, it can be applied to many fields, such as e-commerce, healthcare, finance, etc., to help enterprises and organizations better understand customer needs, market trends, and business operations.

However, despite the wide application prospects of multi-stage frequent data mining methods, existing methods still have some obvious shortcomings. For example, The data was first clustered through the DBSCAN algorithm by Kawtar et al. [7] to discover latent categories and patterns. Then, these clusters are further analyzed using ERT inverse section interpretation to reveal their relationships and structures. By combining the advantages of these two technologies, the information in the data can be mined more comprehensively and deeply. Experimental results show that this method can effectively discover hidden patterns in data and provide more accurate classification results. However, the DBSCAN algorithm has a high computational complexity when clustering large-scale datasets, and the interpretation of ERT cross-sections requires manual parameter settings, which increases processing time and reduces algorithm efficiency. Subrata et al. [8] first clustered the data using the CLS-MMS algorithm to discover different categories and patterns. Then, interesting association rules are found in the data through rare related association rule mining. These association rules can reveal interesting relationships and patterns between categories, providing deeper insights and understanding of the data. Experimental results show that this method can effectively mine frequent items of multi-stage access data. However, the CLS-MMS algorithm and rare correlation rule mining methods require multiple scans of the database, and when processing large-scale datasets, the clustering process and association rule mining have high computational complexity, resulting in a decrease in overall efficiency. Sanz et al. [9] preprocessed the original data set by using fuzzy C-means. Through this operation, the quantitative attribute values can be converted into binary values, and then a fuzzy version of the original data set (consisting of fuzzy records and fuzzy attributes) will be obtained. composition). In addition, a fast rule extraction algorithm based on the fuzzy Apriori algorithm is proposed. This algorithm uses fuzzy clustering to extract fuzzy frequent itemsets from the fuzzy version of the previously obtained original data set to obtain fuzzy association rules. Experimental results show that the proposed new algorithm is superior to the traditional Apriori algorithm in terms of mining time when processing large data sets. However, this method relies on fuzzy C-means clustering to preprocess the original dataset, introducing information loss that may cause certain data features to be blurred, thereby affecting the accuracy of subsequent frequent itemset and association rule mining. Menon et al. [10] aim to mine frequent item sets and interesting relationship patterns in multi-stage access data while protecting sensitive association rules from being leaked. By modifying data in transactional databases, sensitive information is hidden, and mining algorithms are used to discover hidden patterns and association rules. Experimental results show that this method can effectively mine frequent item sets and interesting relationship patterns while protecting sensitive association rules from being leaked. However, this method requires multiple visits to the database during the mining process, and due to data modifications, the mining efficiency is reduced. Magdy et al. [11] use a strategy that combines depth-first and breadth-first to access the search space, use vertical bitmap vector format to store representation itemsets and transaction databases, and use basic pruning strategies, equality pruning strategies, and extended support Equality pruning strategy 1 and extended support equality pruning strategy 2 perform candidate space pruning, and introduce closed candidate sets to incrementally update the mining results to avoid repeated mining of the same frequent item sets. Experimental results show that this method has better performance and accuracy when processing dynamic data streams. However, this method mainly relies on a fixed support threshold for pruning and mining, and fails to fully consider the actual importance of the itemset, resulting in limited accuracy of the mining results.

To illustrate the shortcomings of existing methods and the advantages of this article more intuitively, taking e-commerce as an example: suppose an e-commerce platform wants to analyze user behavior in the three stages of browsing, adding to the shopping cart, and making payments, in order to discover frequently purchased product combinations. Using existing methods, only frequent itemsets based on fixed support thresholds can be mined, while ignoring certain product combinations that are important to the business despite their low frequency of occurrence. In addition, frequent project mining techniques are constantly evolving, with the latest developments including methods based on deep learning. These methods utilize the powerful representation learning ability of deep learning models to automatically extract features from large-scale data and learn complex relationships between itemsets. Although these methods have great potential in theory, they still face many challenges in practical applications, such as high model training costs and limited processing capabilities for large-scale data. In response to the above issues, this article proposes a multi-stage frequent item mining method for accessing data based on association rules. The key contributions of this method include:

1. The Apriori algorithm for association rules has been improved through clustering analysis and the operation of Boolean vectors. This method compresses the matrix by clustering and counting the same transactions, thereby reducing the size of multi-stage database access. It only requires scanning the database once and can directly generate frequent itemsets without generating candidate itemsets, significantly improving algorithm efficiency.
2. Introduce weights into the improved Apriori algorithm and design a weighted support function. By assigning different weights to different itemsets, this method can more accurately reflect the importance and differences of itemsets in practical applications, thereby improving the accuracy of frequent item mining.

Therefore, the method proposed in this article can not only efficiently process large-scale datasets, but also more accurately mine frequent itemsets with practical business significance, providing strong support for enterprise business strategy formulation.

2 Frequent item mining methods for multi-stage access data

2.1 Definition of association rules for frequent item mining of multi-stage access data

An association rule is a rule that expresses a relationship between data items [12], usually in the form of an "if X occurring, then Y is also very likely to happen", recorded as $X \to Y$. Among them, X and Y represent the set of items in the dataset, respectively, and $X \cap Y = \emptyset$(i.e X and Y has no intersection). Association rules play a crucial role in multi-stage access data frequent item mining. By defining key metrics such as support and confidence, and following certain mining steps and strategies, valuable association rules can be effectively mined from large-scale datasets.

Set $I = \{i_1, i_2, \cdots, i_m\}$ be m collection of different multi-stage access data frequent item mining projects. $T = \{t_1, t_2, \cdots, t_n\}$ is a multi-stage access to a database of data-frequent item mining transactions, where each transaction t_j denotes the j-th multi-stage access data frequent item mining transaction of T, i.e., it is a set of multi-stage access to a set of data-frequent item mining projects in I, $t_j \subseteq I$. Each frequent item mining transaction is associated with a unique identifier TID. If for a subset of X in I, there is $X \subseteq t_j$, a frequent term mining transaction shall contain X. An association rule is a rule shaped like "$X \Rightarrow Y$". Implications, in which $X \subseteq I$, $Y \subseteq I$ and $X \cap Y = \emptyset$, $\emptyset$ is the empty set.

The relevant definitions of association rule mining algorithms are as follows:

Definition 1 Degree of support: If the proportion of the frequent item mining transaction $X \cup Y$ in T is S, the association rule $X \Rightarrow Y$ has support S in T, with the following formula:

$$S = \frac{\zeta\left(X \cup Y\right)}{N} \times 100\% \tag{1}$$

Among them, $\zeta\left(X \cup Y\right)$ is the number of transactions supported in the database for multi-stage access $X \cup Y$. N is the total mining number of transactions for frequent items in a multi-stage access database. Multi-stage access database refers to a data structure in which data is organized into multiple stages or levels, allowing users to access and process data at different stages. Taking an e-commerce database as an example, the data is divided into different stages such as user browsing, adding to shopping cart, placing orders, and making payments. Each stage contains relevant data items. The user browsing stage records the product information viewed by the user, while the payment stage records the payment details. This database structure helps analyze user behavior patterns during the purchasing process, but it is more complex to handle as it requires coordination of data access between multiple stages.

Assuming there is an e-commerce database that records 1000 transactions. After analysis, it was found that there were 200 transactions involving milk and bread. So, the support for the association rule 'If you buy milk, you may buy bread' is 20%.

Definition 2 Confidence: If the proportion for X of the multi-stage access frequent item mining transaction in T that also contains Y is C, the association rule $X \Rightarrow Y$ holds with confidence C in T, the formula is as follows.

$$C = \frac{S\left(X \cup Y\right)}{S\left(X\right)} \times 100\% \tag{2}$$

Among them, $S\left(X \cup Y\right)$ is the degree of support for $X \cup Y$. $S\left(X\right)$ is the degree of support for X.

Assuming that out of 300 transactions containing milk, 200 transactions also include bread. So, the confidence level of the association rule 'If you buy milk, you may buy bread' is 67%. This means that 67% of transactions involving the purchase of milk also involve the purchase of bread.

Support and confidence are two units of measure for rules, which reflect their usefulness and certainty, respectively [13]. The mining problem of association rules is to generate all the rules that satisfy the minimum support specified by the user $\min S$ and the minimum confidence level $\min C$ association rules, i.e., the support and confidence of these associations are not less than the minimum support and minimum confidence, respectively.

Definition 3 Term set: A collection of terms is called a term set. For example, {milk, bread} is a itemset. k term set: The set of items containing k terms is called the set of k terms. For example, {milk, bread, eggs} is a set of three items. Frequent Itemset: The set of items that satisfy the minimum support is called frequent itemset. For example, if the minimum support is set to 0.1 (10%), then any itemset with support greater than or equal to 10% is a frequent itemset.

2.2 Multi-stage access data frequent item mining based on Apriori algorithm

In real-world applications, data sets are often large in size and contain thousands of items and transactions. Directly mining association rules on such a data set will involve a huge amount of computation, especially when calculating the support of itemsets. The Apriori algorithm [14] uses an iterative method of layer-by-layer search, starting from frequent 1-item sets and gradually generating frequent k-item sets ($k \geq 2$). A large number of unnecessary candidate item sets are avoided from being generated, and the amount of calculation is reduced by this method. Therefore, mining of multi-stage access data frequent items is carried out in two steps: discovering all frequent item sets of multi-stage access data and mining the corresponding association rules.

This article chooses Apriori as the baseline method mainly because it is a classic algorithm in the field of association rule mining, with the advantages of intuitive and easy to understand implementation. Although there are more efficient new methods such as FP Growth [15] and LCM [16], the extensive application of Apriori algorithm in academia and industry, as well as its applicability in mining frequent items in multi-stage access data, make it an ideal starting point for demonstrating and improving algorithm performance.

A strong association rule means that this rule also satisfies the minimum confidence level under the premise of satisfying the minimum support, and the strong association rule is what needs to be mined in this paper [17]. In this paper, the minimum support threshold is generally denoted as $\min\_S$; the minimum confidence threshold is abbreviated as $\min\_C$.

The mining of Apriori association rules mainly consists of two steps, which are as follows:

Step 1 Discover all frequent itemsets of multi-stage access data.

Step 2 Mining the corresponding association rules [18] based on the obtained frequent item sets.

The Apriori algorithm accesses data frequently through the resulting multi-stage k term sets to generate candidates $(k + 1)$ term sets, which in turn generate frequent item set $(k + 1)$. Firstly, we need to find out the frequent 1-item set of the multi-stage access data, and in this paper, we denote the frequent 1-item set as G_1; and then utilizing the resulting G_1 to produce G_2, i.e., multi-stage access data frequent 2 itemsets; these steps are repeated until no new multi-stage access data itemsets can be generated [19].

In order to show how this a priori property can be used in practice, this paper examines how it can use G_{k-1} to find out G_k, $k \geq 2$. This is accomplished by following two steps in this paper.

(1) Connection step: Firstly, the candidates for multi-stage accessing data k term set need to be found out, which can be passed G_{k-1} self-join to operate. The set of this candidate itemset is denoted as Z_k. Set g_1 and g_2 be the set of multi-stage access data items in G_{k-1}. Notation $g_i\left[j\right]$ denotes the j-th item in g_i. In this paper, let the individual itemsets be arranged in order of size, ensuring that the self-joining operation of $g_i\left[1\right] < g_i\left[2\right] < \cdots < g_i\left[k-1\right]$ is recorded as $g_{k-1} \oplus g_{k-1}$, if there is $(g_1\left[1\right] = g_1\left[1\right]) \wedge \cdots \wedge (g_1\left[k-2\right] = g_2\left[k-2\right]) \wedge (g_1\left[k-1\right] = g_2\left[k-1\right])$, it means that the elements g_1 and g_2 in G_{k-1} is connectable.

(2) Pruning step: in order to compress Z_k, a priori property can be used in the following way. If a multistage access data itemset is infrequent, then all its supersets must also be infrequent [20]. Then this paper can take these infrequent itemsets eliminated from Z_k.

2.3 Multi-stage frequent item mining of access data based on improved Apriori algorithm

The time cost of the Apriori algorithm is mainly that each time a frequent item set is generated, it must traverse the multi-stage access data frequent item mining transaction database and generate a large number of multi-stage access data candidate item sets. In view of the above shortcomings, improvements have been made in this paper. In the improvement, the idea of relational operation of vertical Boolean vectors was adopted, and based on conventional matrix operations, the same frequent item mining transactions that accessed the data warehouse in multiple stages were counted to compress the matrix. The data structure of the matrix was optimized, and rapid multi-processing was achieved. Frequent pattern mining of stage access data [21].

Set $A = \{a_1, a_2, \cdots, a_m\}$ be m collection of different multi-stage visit data frequent item mining projects, task-related multi-stage visit data D is a collection of databases in which each multi-stage access to data is frequently mined for transactions T is the set of items such that $T \subseteq A$. Each multi-stage access data frequent item mining transaction has a unique identifier, called TID.

Definition 4 The vector for each multi-stage access data item A_j is defined as:

$$D_j = (d_{j1} \ d_{j2} \ \cdots \ d_{jn}) \tag{3}$$

Among them, $d_{ij} = \begin{cases} 0, & I_j \notin T_i \\ 1, & I_j \in T_i \end{cases}$; $SC(I_j) = \sum_{i=1}^{n} d_{ij}$.

Definition 5 The matrix of the multi-stage access data item set A is recorded as:

$$D = \begin{pmatrix} D_1 \\ D_2 \\ \vdots \\ D_m \end{pmatrix} = \begin{pmatrix} A_1 \\ A_2 \\ \vdots \\ A_m \end{pmatrix} \begin{pmatrix} d_{11} & d_{12} & \cdots & d_{1n} \\ d_{21} & d_{22} & \cdots & d_{2n} \\ \vdots & \vdots & \cdots & \vdots \\ d_{m1} & d_{m2} & \cdots & d_{mn} \end{pmatrix} \tag{4}$$

Definition 6 The support counts for the two multi-stage access data frequent item mining projects in matrix D is:

$$SC\{A_i, A_j\} = D_i \wedge D_j = \sum_{k=1}^{n} (d_{ik} \wedge d_{jk}) \tag{5}$$

Definition 7 Boolean matrices R of vertical type: For any given multi-stage access to a database D of data-frequent item mining transactions, order: $f: D \to R$, of which: f is a mapping function, $R = f(D) = \left(r_{A_j T_i}\right)_{nm}$, here, $r_{A_j T_i} = \begin{cases} 1 & T_i \neq 0 \\ 0 & T_i = 0 \end{cases}$. As a result, after one scanmulti-stage access to data-frequent item mining transaction databases D, it is mapped to a vertical Boolean matrix by the above definition.
Let each multi-stage access data frequent item mining project a_j have a weight w_j corresponding to it, where $0 \leq w_j \leq 1$, i.e $W(i_j) = w_j$.

Definition 8 For a given multi-stage access data k-itemsets X, its weight is defined as:

$$W(X) = \max\{w_1, w_2, \cdots, w_k\} \tag{6}$$

Among them, k is the number of frequent item mining projects contained in a multi-stage access data item set X, obviously $0 \leq W(X) \leq 1$.

Definition 9 For a certain multi-stage access data k-itemsets X, its weighted support is defined as:

$$wS(X) = W(X) \times S(X) \tag{7}$$

Among them, $S(X)$ is the support for the multi-stage access data item set X.

Definition 10 If a multi-stage access data k-itemsets X its weighted support satisfy $wS(X) \geq w\min S$, it is called weighted multi-stage access data frequent itemsets [22, 23]. Among them, $w\min S$ is the minimum weighted support.

Definition 11 Weighted confidence $wC(X \cup Y)$ of association rules $X \Rightarrow Y$ is:

$$wC(X \cup Y) = \frac{wS(X \cup Y)}{wS(X)} \tag{8}$$

Definition 12 If $X \cup Y$ is a frequent itemset of multi-stage access data, and its weighted confidence is not lower than the minimum weighted confidence $w \min C$, the association rule "$X \Rightarrow Y$" is interesting.

Definition 13 Define Y as a multi-stage access data q-itemsets, $q < k\,(k > 1)$. In the remaining multi-stage access data item set $(A - Y)$, the item with the largest top $(k - q)$ weight is $a_{r_1}, a_{r_2}, \cdots, a_{r_{(k-q)}}$, so the maximum possible value of any k-item set containing the multi-stage access data item set Y is:

$$W\,(Y, k) = \max\,\{w_1, w_2, \cdots, w_m\} \tag{9}$$

If some multi-stage access data k-itemsets X is frequent [24–26], then its support $S\,(X)$ should satisfy the following formula:

$$S\,(X) \geq \frac{w \min S\,(X) \times n}{W\,(X)} \tag{10}$$

Among them, n is the total number of transactions mined for frequent items of multi-phase access data.

Combining formula (9) and formula (10), it can be inferred from this paper: If the multi-stage access data k-item set containing Y is frequent, so the minimum support should be:

$$B\,(Y, k) = \frac{w \min S\,(X) \times n}{W\,(Y, k)} \tag{11}$$

This article calls $B\,(Y, k)$ as the k-support expectations for Y, to ensure that the multi-stage access data k-itemsets set containing Y is likely to be frequent, this paper choose to round up instead of down.

According to formulas (9) and (11), if $w \min S$ and n are given values, it can get $B\,(Y, k)$ as a constant. This paper order $B_{\min} = B\,(Y, k)$ and call it the minimum support expectation.

The specific description of mining frequent items in multi-stage access data based on the improved Apriori algorithm is as follows:

Input: Multi-stage access to data frequent item mining transaction database D; Minimum weighted support threshold $w \min S$.

Output: Frequent itemsets of multi-stage access data in D.

Proceed as follows:

(1) Data preprocessing

a. Scan multi-stage access database D to create a vertical Boolean matrix R, making the vertical Boolean matrix of each column in R compresses and stores one or more multi-stage access data frequent item mining transaction information and there are no duplicate columns (an array can be created to store the weight of each column, that is, the transaction information of the same multi-stage access data frequent item mining item).

b. Specify the number K of clusters to be divided and minimum weighted support $w \min S$. Scan the matrix and calculate the weighted support of all frequent item mining items. If the weighted support of the multi-stage access data frequent item mining item is less than the minimum weighted support, that is $wS < w \min S$ then the frequent item mining project does not play a role in generating frequent 2 item sets. The item can be marked and discarded, that is, the row vector corresponding to the item in the matrix can be deleted. The items retained in the matrix are a set of multi-stage access data frequent 1 item sets.

c. Define the distance $g = \frac{1}{SC}$ between two items, where SC is the number of times the two items appear simultaneously in the transaction record (the more frequent the occurrences at the same time, the smaller the distance, that is, the closer the connection).

d. Random selection K multi-stage access data frequent item mining project used as the initial centroid.

e. According to the distance between each frequent item mining item and each initial centroid, divide the frequent item mining items one by one into the clusters closest to the initial centroid.

f. Whenever the distance from the initial center of mass is less than or equal to $\frac{1}{w \min S}$, all points are assigned to the cluster represented by the centroid, which allows a point to belong to multiple clusters at the same time (ensuring that association rules are not missed). The centroid refers to the center point or average value of a certain category of data in a dataset, which represents the common features or trends of that category of data. In this step, the centroid serves as the representative point of the cluster, allowing data points within a certain range to belong to multiple clusters at the same time, to ensure that association rules are not missed during frequent item mining. This processing method helps to more comprehensively capture the correlation between data items, improving the accuracy and practicality of mining results.

g. Recalculate the centroid of each cluster until the centroid does not change and end the cycle.

h. According to the above steps, the initial frequent item mining item set can be divided into K smaller item set, and accordingly a vertical Boolean matrix R for mining frequent items in multi-stage access data is divided into K smaller vertical Boolean matrix R_K(For the convenience of description, we will use R to represent R_K).

(2) Generate weighted frequent 1-itemsets of multi-stage access data.

a. Find the number of times each multi-stage access data frequent item mining item appears in all transactions. If the support of a 1-item set is greater than or equal to the minimum support required to become a weighted frequent itemset, mark the row vector corresponding to the itemset. At this time, the itemset corresponding to the marked row vector is the weighted frequent 1-itemset of the multi-stage access data.

b. If the support of a 1-item set is less than the minimum support expectation, discard the row vector corresponding to the item set, otherwise save it. Reassemble the saved row vectors into a matrix R^1 in their original order.

c. Respectively sum each column in R^1, and if the sum is less than 2, delete the column vector.

d. The column vectors saved for R^1 are combined into a matrix in the original order to obtain the matrix R^2.

(3) Generate weighted frequent 2-itemsets of multi-stage access data. Calculate on each line of R^2, and calculates the number of times two items appear simultaneously in all multi-stage access data frequent item mining transactions. If the support of a 2-item set is greater than or equal to the weighted frequent item set, The required minimum

support,), then mark the row vector corresponding to the item set. At this time, the item set corresponding to the marked row vector is the multi-stage access data weighted frequent 2-item set. If the support of a 2-itemset is less than the minimum support expectation, discard the row vector corresponding to the itemset, otherwise save it. Reassemble the saved row vectors into a matrix R^3 in their original order. Respectively sum each column in R^3, and if the sum is less than 2, delete the column vector.The column vectors saved by R^3 are combined into a matrix in the original order to obtain the matrix R^4.

(4) Generate weighted frequency of multi-stage access data K-item set. The operation steps are the same as step 3.

(5) Convert all marked row vectors into corresponding multi-stage access data weighted frequent item sets and output them to complete multi-stage access data frequent item mining.

The significance of introducing the concept of weighted mining in current research is to more accurately reflect the importance and differences of different itemsets in practical applications, thereby improving the accuracy and practicality of frequent item mining. Although setting project weights can improve the specificity of the method, designing a reasonable weight allocation strategy can reduce domain dependence to a certain extent, ensuring the universality and adaptability of the method in different application scenarios.

To demonstrate the principle of the proposed improved Apriori algorithm, a specific example of mining frequent items in multi-stage access data is presented. Assuming there is an e-commerce database that records users' purchasing behavior at different stages, including browsing, adding to the shopping cart, placing orders, and making payments. In this database, each user's purchasing behavior is treated as a transaction, which includes the product items that the user browsed and purchased at various stages. Now, by mining the association rules between these product items, we can understand which products are often purchased together. Using the improved Apriori algorithm, first scan the database to establish a vertical Boolean matrix that can compress and store the same transaction information and calculate the weighted support for each product item. Next, frequent itemsets are selected based on weighted support, and candidate itemsets are generated using these frequent itemsets. Through continuous iteration and pruning steps, all weighted frequent itemsets are discovered, and ultimately association rules that satisfy the minimum weighted support and weighted confidence are mined. For example, association rules such as' if a user purchases milk, they are likely to also purchase bread 'may be found, which is very useful for businesses to develop marketing strategies and optimize product display layouts.

3 Experimental analysis

Taking the multi-stage access database of an electronic library system as the experimental object, the multi-stage access database contains access data from the identity authentication stage, data request stage, data transmission stage and data query stage from September 2010 to August 2022. 15 e-books with the most access times were selected for a multi-stage access data frequent item mining experiment in order to reduce the amount of data. Statistical attributes include: e-book MAC number, title, publisher, publication date and number of visits. Some information on the multi-stage access data of these books is shown in Table 1.

Using the method described in this article, the frequent items of multi-stage access to the database of the electronic library system are mined. Before introducing the mining results, an example user ID dataset is provided in the experiment, as shown in Table 2.

The results of mining frequent items in the multi-stage access data are shown in Table 3.

The user ID in Table 3 represents the data in the identity authentication stage, and the book title represents the data in the query stage. The weighted support of each frequent item is higher than the minimum support expectation, which indicates that the mined frequent items meet the expectation as their weighted support exceeds the preset value. These frequent items are in line with the requirements of data mining. In Table 3, it can be seen that the confidence of each frequent item is relatively high, which means that the accuracy and credibility of these frequent items are relatively high. These frequent items can effectively reflect the user's experience in the multi-stage access process. Behavioral patterns, by analyzing frequent items in multi-stage access data, can help this paper better understand user behavior and provide a basis for optimizing the user experience of the electronic library system. In summary, the method in this paper is effective in mining frequent items in multi-stage access data, and can effectively reflect the user's behavior pattern in the multi-stage access process.

In order to observe the correlation between users and books more intuitively, the correlation strength diagram between users and books was generated based on the mined frequent items. The correlation strength between some users and books is shown in Fig. 1. The thicker the connection in the figure, the stronger the correlation between the two; the thinner the connection, the weaker the correlation. The link in the picture is the number of times the user and the book appear at the same time.

It can be found from Fig. 1 that the strongest connection between User241, User229, User207 and Shadow's complaint is the thickest, indicating that the strongest correlation

Table 1 Partial information on multi-stage access data

Serial number	Book MAC number	Title proper	Publisher	Publication date	Number of visits
1	0046974	Shadow accusation	Takashi Tsuchiya	1961	837
2	0002827	Quadrillion SUNS	Rudolf Kippenhahn	2021	759
3	0058299	Mechanical Design 2nd edition	Yasaki, Gu Daqiang	2016	735
4	0069919	Bow puzzle	Yriel Ranwill	1892	657
5	0059974	The old man in the corner	Baroness Orsiz	1901	553
6	00717	Mechanical parts design	Pan Chengyi	2021	545
7	0024752	On the hoof	Yu Hua	1993	431
8	0001605	Ordinary world	Lu Yao	1986	423
9	0046077	Mechanical design	Men Yanzhong	2018	411
10	0057807	The story of art	Gombrich	2008	402
11	0048752	Tourism anthropology	Nelson Winter	unknown	387
12	00605	Origin of species	Charles Darwin	1859	354
13	00504	Tourism philosophy	Hans Funk	unknown	306
14	00716	Education through the arts	Herb Reed	1993	287
15	00817	Basis of science	Xia Yulong	1983	255

Table 2 Example user ID dataset

User ID	User name	Registration date	Description
User058	Alice	2015-03-15	Frequent access to literary books and high interest in 'The Ordinary World'
User121	Bob	2016-07-22	Mechanical engineering students often consult books related to mechanical design
User228	Charlie	2018-11-05	Biology enthusiast, interested in evolution theory and long-standing scientific works
User132	Diana	2017-04-28	Likes science fiction and futuristic works
User458	Edward	2020-09-10	Frequently read foreign literary classics
User241	Francesca	2019-02-14	Legal background, focusing on novels with reasoning and legal themes
User302	George	2014-12-08	Educators focus on the application of art in education
User117	Hannah	2017-08-21	Indulge in puzzle games and intellectual challenges, enjoy books like 'Bow Puzzle'
User213	Ivan	2021-01-15	Like User132, follow Quadrillion SUNS
User229	Jack	2022-05-30	Pay attention to historical novels and suspense mystery books
User173	Katherine	2016-11-07	Indulging in basic scientific theories and frequently reading books on scientific principles
User207	Lawrence	2020-02-20	Similar reading preferences to User241 and User229, liking historical suspense novels
User186	Maria	2019-07-03	Art history and aesthetics enthusiasts with in-depth research on art theory
User193	Nicole	2018-03-25	Novels that focus on animals and rural life themes
User215	Oliver	2021-09-12	Mechanical engineer with a continuous demand for mechanical design books

Table 3 Mining results of frequent items in multi-stage access data

Frequent items (User → title)	Weighted support/%	Minimum support expectation/%	Weighted confidence
User058→Ordinary world	1.073	1	100
User121→Mechanical parts design	1.073	1	100
User228→Origin of species	1.336	1	100
User132→Quadrillion SUNS	1.281	1	100
User458→The old man in the corner	1.118	1	100
User241→Shadow accusation	1.373	1	100
User302→Education through the arts	1.173	1	100
User117→Bow puzzle	1.289	1	98.98
User213→Quadrillion SUNS	1.299	1	98.63
User229→Shadow accusation	1.363	1	96.76
User173→Basis of science	1.263	1	96.76
User207→Shadow accusation	1.399	1	96.17
User186→The story of art	1.008	1	95.42
User193→On the hoof	1.045	1	95.42
User215→Mechanical design	1.045	1	95.42

between these three users and Shadow's complaint; User132, User213 and the correlation strength between Hundreds of Billions of Suns is second only to the correlation strength between User241, User229, User207 and Shadow's complaints, indicating that the correlation strength between these two users and Hundreds of Billions of Suns is relatively strong; the correlation between other users and different books The correlation strength is weak. Figure 1 can help electronic library system managers understand each user's behavior pattern during the multi-stage access process.

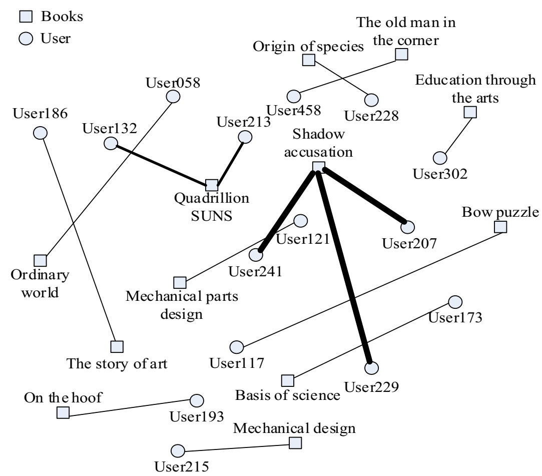

Fig. 1 Part of the strength of the association between users and books

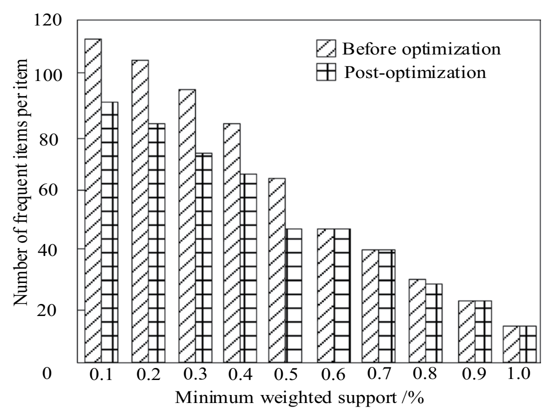

Fig. 2 Changes in the number of frequent items with different minimum weighted support degrees

The number of frequent items produced by this method under different minimum support conditions is analyzed. The number of transactions in the data set remains unchanged. Analyze the changes in the number of frequent item sets as the minimum weighted support increases. The analysis results are shown in Fig. 2.

It can be seen from Fig. 2 that as the minimum weighted support continues to increase, the number of generated frequent items gradually decreases. When the weighted support is small, after the optimization of this method, the number of frequent items generated is significantly different from that before optimization. The number of frequent items after optimization is significantly lower than the number of frequent items before optimization. When the minimum weighted support increases to 0.6%, the number of frequent items generated before and after the optimization,of this method tends to be consistent. The main reason is that when the weighted support is very small, there are more frequent items that meet the conditions, and the running time of the method will be correspondingly longer. However, as the minimum weighted support increases, there will be fewer and fewer frequent items and association rules that meet the conditions and tend to be relatively stable.

The multi-stage access data frequent items mined by this method when the minimum support expectation is 1.2% are analyzed. The mining results are shown in Fig. 3.

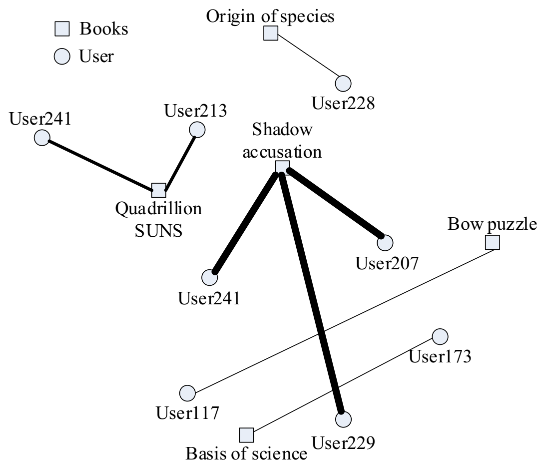

Fig. 3 Frequent item mining results of multi-stage access data when the minimum support expectation is 1.2%

Table 4 Pearson coefficient analysis results of frequent item mining of multi-stage access data

Serial number	Title proper	Pearson coefficient
1	Shadow accusation	0.992
2	Quadrillion SUNS	0.987
3	Mechanical design 2nd edition	0.997
4	Bow puzzle	0.981
5	The old man in the corner	0.986
6	Mechanical parts design	0.996
7	On the hoof	0.995
8	Ordinary world	0.993
9	Mechanical design	0.989
10	The story of art	0.987
11	Tourism anthropology	0.998
12	Origin of species	0.981
13	Tourism philosophy	0.983
14	Education through the arts	0.995
15	Basis of science	0.984

As can be seen from Fig. 3, a total of 8 frequent items in the multi-stage access data were mined. Compared with Fig. 1, it can be seen that frequent items with weighted support lower than 1.2% have been removed. This shows that the number of frequent items that meet the conditions of multi-stage access data is significantly reduced at this time, which is basically consistent with the changing trend of the number of frequent items at different minimum weighted supports in Fig. 2.

The Pearson coefficient is used to measure the mining accuracy of frequent items in multi-stage access data using this method. The closer the Pearson coefficient is to 1, the higher the accuracy of mining frequent items in multi-stage access data. The method in this article to mine multi-stage access related to 15 books in Table 1 is analyzed. The Pearson coefficient when the data is a frequent item, the analysis results are shown in Table 4.

As can be seen from Table 4, for different books, the Pearson coefficient of the frequent items of multi-stage access data mined by this method is relatively close to 1, and the average Pearson coefficient is 0.9896. According to the frequent items mined by this method, it can be found that these frequent items reflect the user's access patterns and behavioral preferences for different books in the multi-stage access process. These pieces of information are crucial for electronic library systems as they can help to better understand users' needs and preferences for different books, thereby providing users with more personalized and accurate book recommendation services. In addition, these frequent items can also be used to evaluate the quality and popularity of books, and provide references for book procurement, typesetting and promotion.

The comprehensive evaluation function value is used to evaluate the multi-stage access data frequent item mining effect of this method. The higher the value, the better the multi-stage access data frequent item mining effect of this method. The DBSCAN clustering mining method studied by Kawtar et al., Subrata et al. The CLS-MMS mining method studied by Sanz et al., the fuzzy association rule mining method studied by Sanz et al., the modified transactional database mining method studied by Menon et al., and the closed candidate mining method studied by Magdy et al. are used as comparison methods for the method in this article. The comprehensive evaluation function method for mining frequent items in multi-stage access data using six methods with different minimum weighted supports is analyzed, and the analysis results are shown in Fig. 4.

From Fig. 4, it can be seen that the y-axis represents the comprehensive evaluation function value, which is an indicator that comprehensively considers mining efficiency, accuracy, stability, and other possible factors (memory usage, computational complexity, etc.), and is used to comprehensively evaluate the effectiveness of mining methods. Specifically, the higher the comprehensive evaluation function value, the better the performance of the method in mining frequent items in multi-stage access data. From the graph, it can be seen that as the minimum weighted support degree increases, the comprehensive evaluation function values of all methods show an upward trend. However, a fast improvement speed is exhibited by the method proposed in this paper in the early stage, demonstrating its efficient mining ability, and a better comprehensive evaluation function value is always maintained by it than other comparative methods throughout the entire process. This result not only highlights the stability and superiority of our method under different minimum weighted support degrees, but also provides valuable

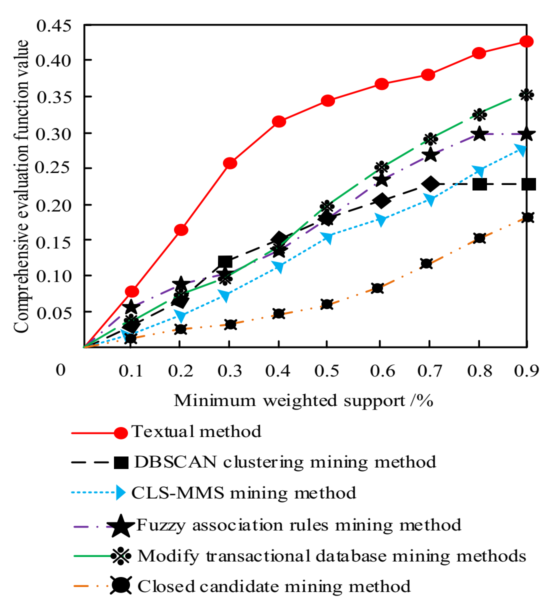

Fig. 4 Comprehensive evaluation function analysis results

Table 5 Performance evaluation results of different frequent item mining methods

Data subset size	Method	Computational complexity (s)	Memory usage (MB)	Run time performance (rating)
Small	Textual method	123	45	9.2
	DBSCAN clustering mining method	187	68	7.8
	CLS-MMS mining method	154	56	8.4
	Fuzzy association rules mining method	192	72	7.5
	Modify transactional database mining methods	168	62	8.0
	Closed candidate mining method	141	52	8.8
Medium	Textual method	345	128	8.8
	DBSCAN clustering mining method	567	214	6.5
	CLS-MMS mining method	489	182	7.2
	Fuzzy association rules mining method	612	236	6.0
	Modify transactional database mining methods	523	198	6.8
	Closed candidate mining method	421	156	7.6
Large	Textual method	867	384	8.0
	DBSCAN clustering mining method	1845	728	4.5
	CLS-MMS mining method	1423	546	5.5
	Fuzzy association rules mining method	1968	784	4.0
	Modify transactional database mining methods	1634	632	5.0
	Closed candidate mining method	1123	428	6.0

reference for research and application in related fields by exploring the impact of parameter sensitivity on mining results.

In order to further evaluate the performance of the multi-stage access data frequent item mining method proposed in this article, the multi-stage access database of an electronic library system was continued to be used as the experimental object. Different sized subsets of data were selected, and six methods were applied separately, and the computational complexity, memory usage, and runtime performance were recorded. The experimental results are shown in Table 5.

According to Table 5 analysis, relatively stable performance is exhibited by the Textual method in terms of computational complexity, memory usage, and runtime performance on subsets of data of different sizes. In a small dataset, the computational complexity of the Textual method is 123s, the memory usage is 45MB, and the runtime performance score is 9.2. Although the memory usage is not achieved, good computational complexity and runtime performance are maintained. As the size of the dataset increases, the computational complexity and memory usage are gradually increased by the Textual method, but the growth rate is more gradual compared to other methods. In a medium-sized dataset, its computational complexity is 345s, memory usage is 128MB, and runtime performance score is 8.8, still maintaining a high performance level. In large datasets, although the computational complexity of the Textual method reaches 867 seconds and the memory usage reaches 384MB, which is significantly increased compared to small datasets, its runtime performance score is 8.0 compared to the other five methods, still at a high level, indicating that the Textual method can maintain relatively stable performance when processing large-scale data. Overall, the Textual method performs outstandingly in terms of computational complexity and runtime performance, especially when the dataset size is large, its performance stability is more significant.

4 Conclusion

An efficient and practical data mining technique is introduced in this article—the multi-stage frequent item mining method based on association rules. Frequent items in multi-stage access data can be effectively mined, user behavior patterns and access paths can be revealed, and large-scale data can be handled with ease by it. Through association rule inference, valuable information in the data can be excavated to assist decision-making. Wide application potential has been shown by it in fields such as e-commerce, book systems, and recommendation systems, helping enterprises to deeply understand user needs, optimize products and services, and thereby improve marketing effectiveness.

Three core areas are focused on by the future research direction: firstly, optimizing algorithm performance, aiming to explore algorithm optimization strategies in depth to improve the efficiency and accuracy of processing large-scale data; The second is technology integration, which plans to combine association rules with clustering, classification and other technologies, aiming to expand the depth and breadth of data analysis and mining functions; The third is deep learning applications, dedicated to using deep learning techniques to parse multi-stage access data in order to reveal more complex user behavior patterns.

In the future, personalized recommendation services can not only be achieved, and user experience can be improved by mining user behavior patterns during multi-stage visits using this method; User access paths and behavioral preferences can also be accurately analyzed, advertising strategies can be optimized, and advertising effectiveness can be enhanced by it; In addition, in the field of network security, abnormal access patterns can be effectively identified, potential network attack behaviors can be timely detected, and strong support can be provided for network security protection by it.

In summary, broad application prospects and value are possessed by the multi-stage frequent item mining method based on association rules, and more breakthroughs and innovations will be brought to the field of data mining through future research and development.

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